

Causal Inference

Lecture 12

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- 1 Definition and identification
 - Estimation under (conditional) exchangeability
- 2 Instrumental variables
 - Estimation with IVs
 - Sensitivity Analysis for Invalid Instruments
- 3 Mediation analysis
 - Estimation in mediation analysis
- 4 Final remarks and prospective study

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Identification in causal inference

- In causal inference, identification refers to the question of whether the causal parameter (or effect) can be uniquely determined from the observed data.
- This is crucial because causal parameters are defined using potential outcomes – at least one of which is counterfactual (and therefore unobserved).
- Unlike associations in classical statistics, causal effects cannot be directly computed from unadjusted data due to the presence of possible unmeasured confounders.

Additional assumptions

Identification requires additional assumptions and tools to “recover” the latent causal effects from the observed data.

Tools for Identification

- Directed Acyclic Graphs (DAGs): Visual representations of causal relationships.
- Conditional Exchangeability: The assumption that, given a set of observed covariates, the treatment assignment is independent of the potential outcomes.
- Unconditional Exchangeability: Occurs when treatment is randomly assigned, so the groups are comparable without further adjustment.
- Colliders and Confounders: Variables that, respectively, distort or confound the observed association between treatment and outcome.
- d-separation: A criterion used in DAGs to decide whether a set of variables blocks all non-causal (or causal but irrelevant) paths, helping to identify valid adjustment sets.

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Estimation

Once a causal parameter is identified, various estimation methods can be applied, each with different properties. While identification is a nonparametric issue, estimation is carried out within the framework of classical statistical inference. Common estimation methods include:

Standardization and IPW

- **Standardization (or Stratification):** Directly uses the conditional exchangeability assumption by estimating outcome means within strata defined by covariates.
- **Inverse Probability Weighting (IPW):** Weights the observed data to create a pseudo-population in which treatment is independent of confounders, mimicking unconditional exchangeability.
- **Doubly Robust Methods:** Combine standardization (an outcome model) with IPW (an exposure or propensity score model). These estimators are “doubly robust” in the sense that they remain statistically consistent if either the outcome model or the exposure model is correctly specified, even if the other is misspecified.

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Relaxing assumptions

- When the assumption of conditional exchangeability (with respect to the measured covariates) is implausible, we can use an instrumental variable (IV) to identify and estimate the causal effect in the presence of unmeasured confounders.
- An IV provides a source of variation in the exposure that is independent of the unmeasured confounders.
- We no longer require conditional exchangeability with respect to measured covariates because the IV serves as a “stand-in” for random assignment. It allows identification of the causal effect even when unmeasured confounders are present.

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Two common estimation approaches

- Two-Stage Least Squares (TSLS):
This method is typically used with linear models and involves:
 - 1 Predicting the exposure using the IV (first stage).
 - 2 Estimating the causal parameter by regressing the outcome on the predicted exposure (second stage).
- G-Estimation:
This method can be applied to both linear and nonlinear models. It seeks the value of the causal parameter that “validates” the instrument by ensuring that the instrument is independent of the potential outcomes after adjusting for the measured covariates.

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Sensitivity Analysis for Invalid Instruments

Because the assumptions underlying valid IV are often unverifiable and heavily depend on subject-matter knowledge, it is important to assess the robustness of the results.

Sensitivity analyses for invalid instruments allow us to:

- Identify and quantify potential violations of the IV assumptions.
- Adjust the estimating equations and estimators according to the violation structure.

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Understanding How the Exposure Affects the Outcome

- Mediation analysis aims to uncover the mechanisms and specific pathways through which a causal effect operates.
- The overall strategy is like other parts of the course: we first address definition and identification, and then we focus on estimation of the direct and indirect effects.
- We decompose the total effect of an exposure on an outcome into its natural direct and indirect (mediated) effects.
- This decomposition requires the notion of nested potential outcomes.

No unmeasured confounders and cross-world independence

- Identification in mediation analysis requires stronger assumptions than for total effects (as discussed in Section 1). In particular:
 - No Unmeasured Confounding:
Not only must there be no unmeasured confounders for the exposure–outcome and mediator–outcome relationships, but we also need to assume no unmeasured confounding for the exposure–mediator relationship.
 - Sequential Ignorability and Cross-World Independence:
Even with randomization of the exposure, randomizing the mediator (or conditioning on the mediator) does not guarantee identification of the direct effect. One must also assume cross-world independence, which is a part of the sequential ignorability assumption. This assumption requires that the potential mediator values under different exposure levels are independent of the potential outcomes under any exposure, given observed covariates. This cross-world independence assumption is inherently untestable.
 - Mediation Formula:
Under these strong assumptions, the mediation formula can be used to identify and compute the natural direct and indirect effects.

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Imputation-Based and Weighting-Based Methods

Once identification is established, the causal mediation effects can be estimated using two main methods:

- **Imputation-Based Methods (Standardization):**
These methods involve fitting an outcome model and using it to impute counterfactual outcomes, much like standardization methods used for total effects.
- **Weighting-Based Methods (IPW):**
Similar to inverse probability weighting in total effect estimation, these methods use a model for the exposure (or mediator) to create weights that adjust for confounding in the estimation of mediation effects.

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Approach of this Course

- This course presented a statistical, classical approach to causal inference.
- While other courses may emphasize machine learning techniques and algorithms, the fundamental principles of causal inference remain intact and relevant.
- There are many additional, fundamental topics that were not covered in this course. Examples include difference-in-differences, synthetic control methods, causal discovery, causal survival analysis, and more.
- We hope that this course has equipped you with the essential tools to continue exploring and self-learning these and other topics.